Cover letter for JMLR MLOSS submission

This submission is intended for the JMLR Machine Learning Open Source Software (MLOSS) track.

Manuscript title: *SOPHI: An Open End-to-End Camera–LiDAR Planning Stack with Sparse-Convolution Inference for NAVSIM*

1. Overlap with prior publication. This manuscript describes open-source software for training, evaluating, and deploying a camera–LiDAR fused BEV planner on NAVSIM, including an open sparse-convolution LiDAR runtime that replaces NVIDIA CUDA-BEVFusion's closed `libspconv` path, multi-head ONNX/TensorRT export (planning + detection + BEV segmentation), and a documented reproducibility cookbook. No portion has appeared in a journal. Related public artifacts (CUDA-BEVFusion, Hydra-MDP/GTRS, NAVSIM, `spconv`) are cited and distinguished from our integration and deployment contributions.

2. Co-author consent. This is a single-author submission. The author confirms the manuscript and consents to its review by JMLR.

3. Conflicts of interest. None declared.

4. Open-source license. Apache License 2.0 (see `LICENSE` in the repository and submission archive).

5. Project URL and version.

- Repository: `https://github.com/atharvsharma1998/hydra-mdp`
- Reviewed version tag: `v0.1.0-mloss`
- Companion cookbook: `jmlr/REPRODUCIBILITY.md`

6. Evidence of user community. The project is public at the URL above with Issues enabled, Apache-2.0 licensing, a tagged release `v0.1.0-mloss`, and end-user-end documentation (`README.md`, `INSTALL.md`, `docs/REPRODUCIBILITY.md`) covering train / PDM eval / ONNX / TensorRT / C++ parity. As a young single-maintainer release, external stars and forks are still accumulating; the submission emphasizes reproducibility artifacts (self-contained source archive, validated navtest PDM score, Python–C++ parity) so reviewers and new users can adopt and extend the stack. We welcome Issues and PRs.

7. Suggested action editors (no COI):

- Editors covering open-source ML software, robotics / autonomous driving systems, or efficient deep learning deployment (please assign per MLOSS board expertise).

8. Suggested reviewers (no COI) — suggestions by expertise:

- Researchers familiar with BEVFusion / multi-sensor BEV perception.
- Maintainers or heavy users of `spconv` / sparse 3D convolution runtimes.
- NAVSIM / learned planning (Hydra-MDP, GTRS, PDM scoring) practitioners.
- TensorRT / ONNX deployment systems reviewers.

(List concrete names and emails in the portal form when submitting.)

9. Keywords: open source software; autonomous driving; bird's-eye view; sparse convolution; TensorRT; NAVSIM

10. Source code. Source code is included in the submission archive and is publicly available at the project URL under Apache-2.0 (tag `v0.1.0-mloss`). The archive excludes multi-GB datasets and optional large checkpoints; download instructions and a public checkpoint path are documented in `REPRODUCIBILITY.md`.

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11. Funding disclosure. None.

12. Competing interests. None declared.

13. Proprietary dependencies (justification). The documented C++ inference path requires NVIDIA CUDA and TensorRT, which are proprietary but widely used in the ML systems community (analogous to other accepted GPU MLOSS packages). The LiDAR SCN path does **not** require NVIDIA's closed `libspconv`; it uses open `spconv` 2.x. NAVSIM data is obtained separately under the benchmark's license.